What is NAT (Network Address Translation)

Every IP-capable device needs an IP address. With the introduction of small office, home office (SOHO) networks, this would seem to imply that whenever a SOHO wants to install a LAN to connect multiple machines, a range of addresses would need to be allocated by the ISP to cover all of the SOHO's IP devices (including phones, tablets, gaming devices, IP TVs, printers and more).

If the network grew bigger, a larger block of addresses would have to be allocated. But what if the ISP had already allocated the continuous portions of the SOHO network's current address range? And what typical homeowner wants (or should need) to know how to manage IP addresses in the first place? Fortunately, there is a simpler approach to address allocation that has found increasingly widespread use in such scenarios: network address translation (NAT).

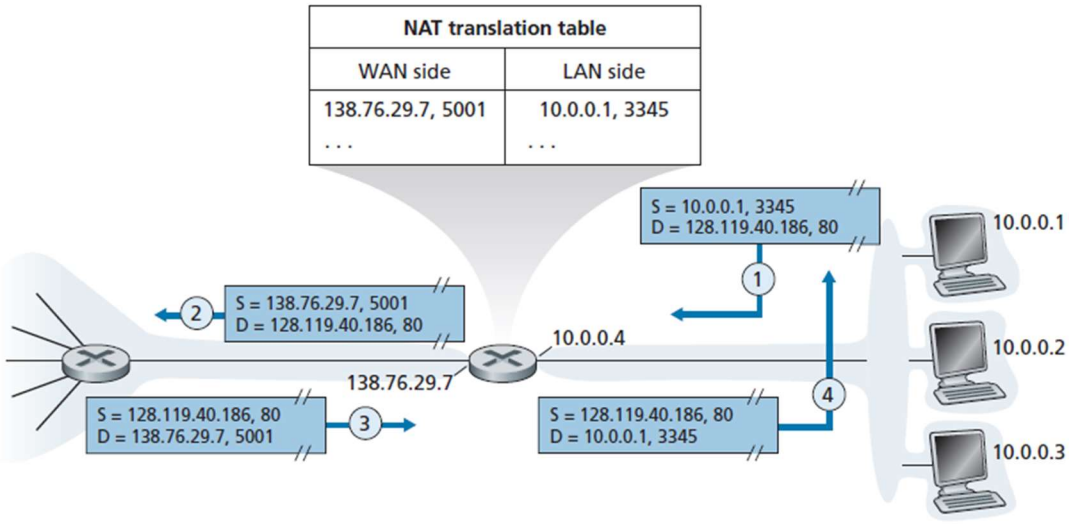


Figure 4.25 ♦ Network address translation

Figure 4.25 shows the operation of a NAT-enabled router. The NAT-enabled router, residing in the home, has an interface that is part of the home network on the right of Figure 4.25.

Q1. What are public and private IP addresses?

A realm with private addresses refers to a network whose addresses only have meaning to devices within that network. There are hundreds of thousands of home networks, many using the same address space, e.g. 10.0.0.0/24. Devices within a given home network can send packets to each other using 10.0.0.0/24 addressing. However, packets forwarded *beyond* the home network into the larger global Internet clearly cannot use these addresses (as either a source or a destination address) because there are hundreds of thousands of networks using this block of addresses. That is, the 10.0.0.0/24 addresses can only have meaning within the given home network.

The NAT-enabled router does not look like a router to the outside world. Instead the NAT router behaves to the outside world as a single device with a single IP address. In Figure 4.25, all traffic

leaving the home router for the larger Internet has a source IP address of 138.76.29.7, and all traffic entering the home router must have a destination address of 138.76.29.7. In essence, the NAT-enabled router is hiding the details of the home network from the outside world.

Q2. Where the home network computers get their addresses and where the router gets its single IP address?

Often, the answer is the same—DHCP! The router gets its address from the ISP's DHCP server, and the router runs a DHCP server to provide addresses to computers within the NAT-DHCP-router-controlled home network's address space.

What's Inside a Router?

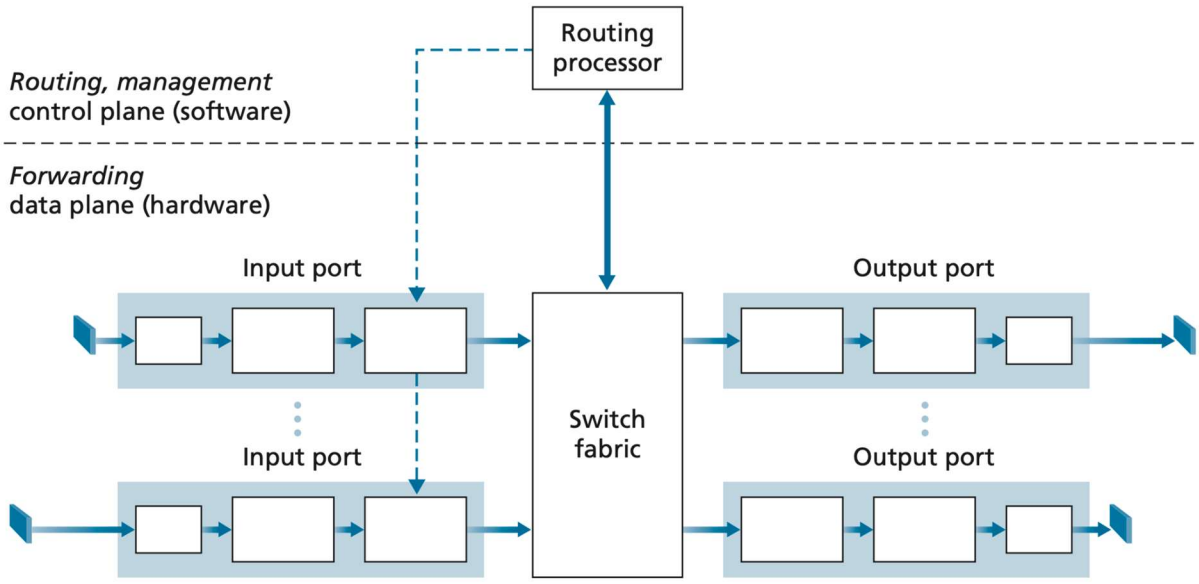


Figure 4.4

1. **Input ports.** An input port performs several key functions. It performs the physical layer function of terminating an incoming physical link at a router; this is shown in the leftmost box of an input port and the rightmost box of an output port in Figure 4.4. An input port also performs link-layer functions needed to interoperate with the link layer at the other side of the incoming link; this is represented by the middle boxes in the input and output ports. Perhaps most crucially, a lookup function is also performed at the input port; this will occur in the rightmost box of the input port. It is here that the forwarding table is consulted to determine the router output port to which an arriving packet will be forwarded via the switching fabric. Control packets (for example, packets carrying routing protocol information) are forwarded from an input port to the routing processor. Note that the term "port" here—referring to the physical input and output router interfaces—is distinctly different from the software ports associated with network applications and sockets

discussed in Chapters 2 and 3. In practice, the number of ports supported by a router can range from a relatively small number in enterprise routers, to hundreds of 10 Gbps ports in a router at an ISP's edge, where the number of incoming lines tends to be the greatest.

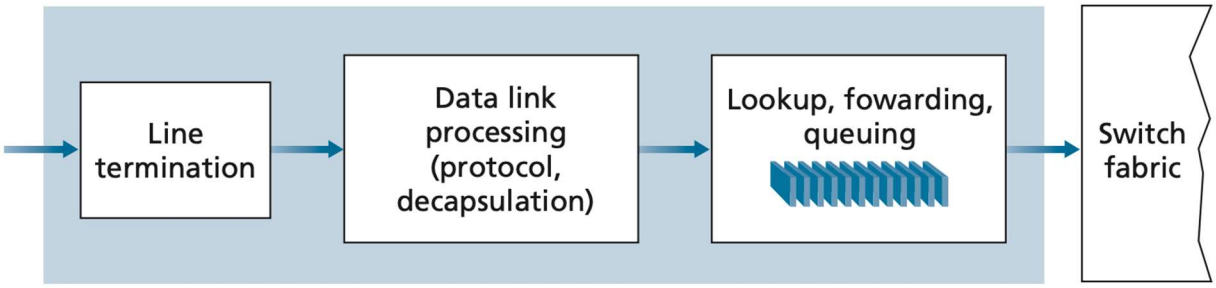


Figure 4.5 ♦ Input port processing

2. **Switching fabric**. The switching fabric connects the router's input ports to its output ports. This switching fabric is completely contained within the router—a network inside of a network router!
3. **Output ports.** An output port stores packets received from the switching fabric and transmits these packets on the outgoing link by performing the necessary link-layer and physical-layer functions.

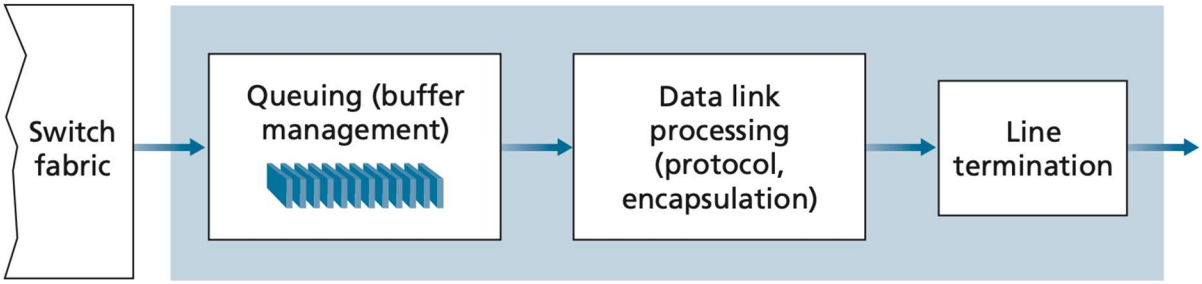
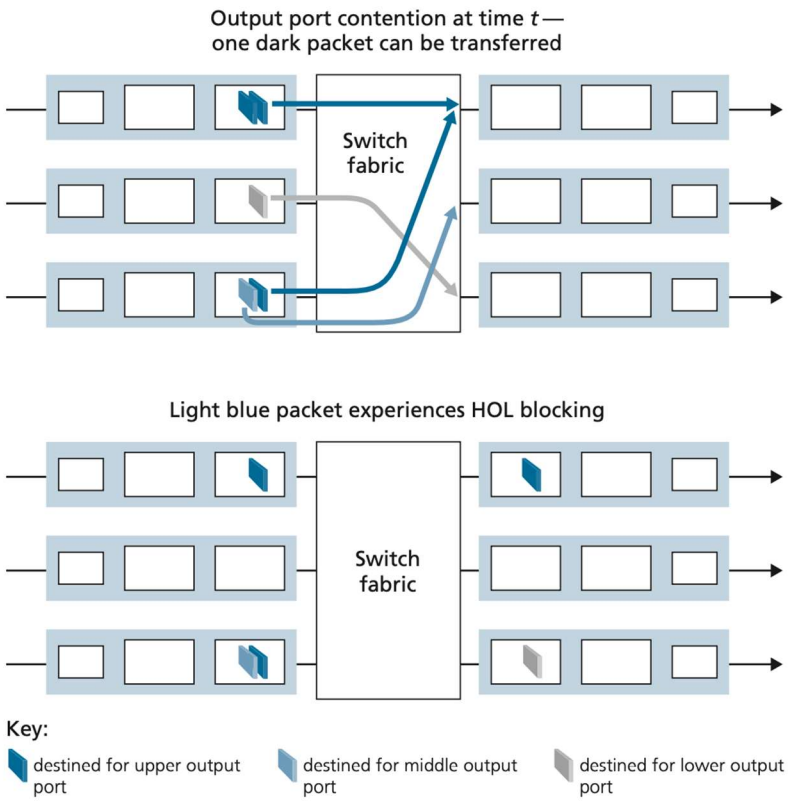


Figure 4.7 ♦ Output port processing

4. **Routing processor.** The routing processor performs control-plane functions. In traditional routers, it executes the routing protocols, maintains routing tables and attached link state information, and computes the forwarding table for the router.

A router's input ports, output ports, and switching fabric are almost always implemented in hardware. To appreciate why a hardware implementation is needed, consider that with a 100 Gbps input link and a 64-byte IP datagram, the input port has only 5.12 ns to process the datagram before another datagram may arrive.

Q3. If the router does not keep up to speed with the datagrams what do you think will happen?



jure 4.8 ♦ HOL blocking at and input-queued switch

Q4. Explain the picture and from memory Head Of Line (HOL) blocking as shown in the diagram.

Longest Prefix Matching in forwarding table

Let's now consider the "simplest" case that the output port to which an incoming packet is to be switched is based on the packet's destination address. In the case of 32-bit IP addresses, a simple implementation of the forwarding table would have one entry for every possible destination address. Since there are more than 4 billion possible addresses, this option is totally out of the question.

As an example of how this issue of scale can be handled, let's suppose that our router has four links, numbered 0 through 3, and that packets are to be forwarded to the link interfaces as follows:

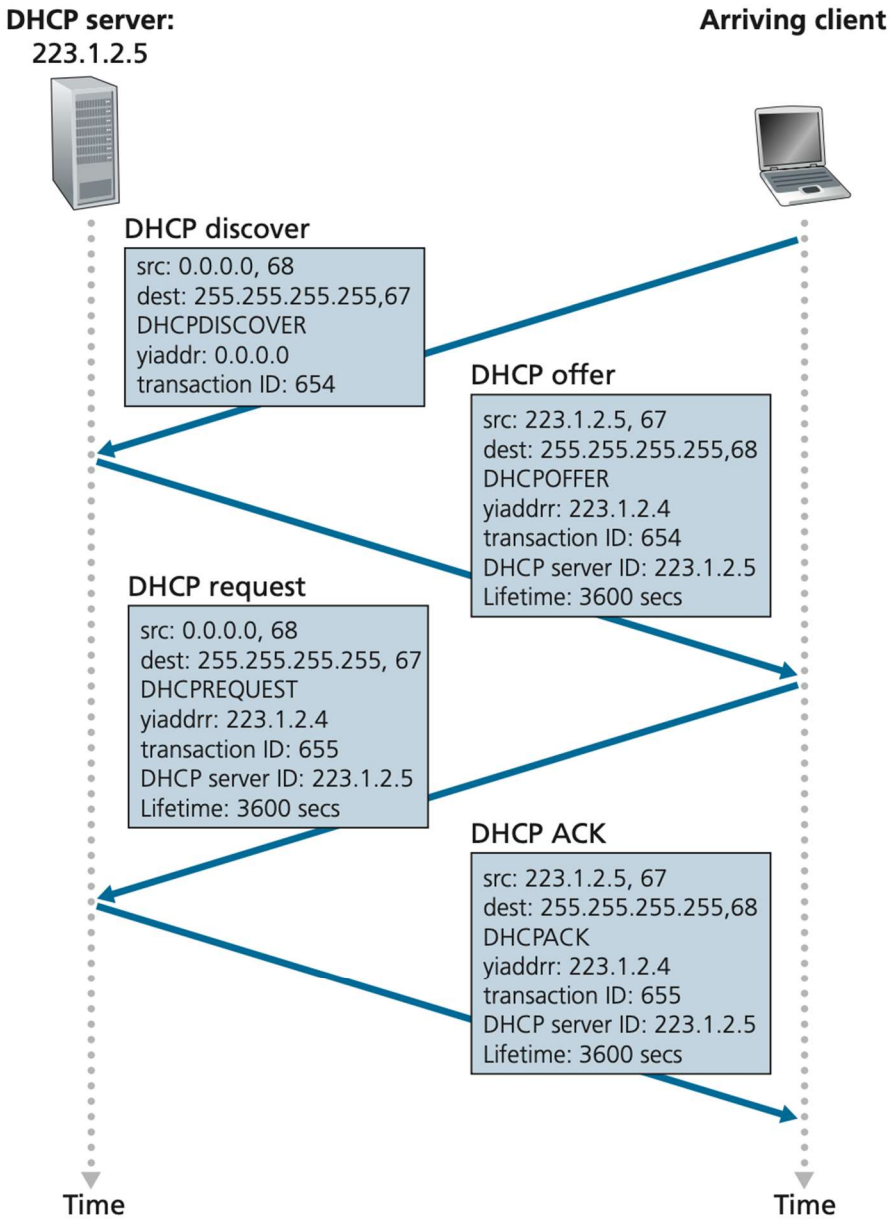
Destination Address Range	Link Interface
11001000 00010111 00010000 00000000 through 11001000 00010111 00010111 11111111	0
11001000 00010111 00011000 00000000 through 11001000 00010111 00011000 11111111	1
11001000 00010111 00011001 00000000 through 11001000 00010111 00011111 11111111	2
Otherwise	3

Clearly, for this example, it is not necessary to have 4 billion entries in the router's forwarding table. We could, for example, have the following forwarding table with just four entries:

Prefix	Link Interface
11001000 00010111 00010	0
11001000 00010111 00011000	1
11001000 00010111 00011	2
Otherwise	3

Q3: How much time do you think this lookup will take while performing matching? What kind of matching strategies are used inside router?

Dynamic Host Configuration Protocol (DHCP)



gure 4.24 ♦ DHCP client-server interaction

Given this, the DHCP client creates an IP datagram containing its DHCP discover message along with the broadcast destination IP address of 255.255.255.255 and a "this host" source IP address of 0.0.0.0. The DHCP client passes the IP datagram to the link layer, which then broadcasts this frame to all nodes.